TAMAR NOSELIDZE

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Quantum Information Master's student at Sorbonne University combining a strong mathematical foundation with machine learning, convex optimisation, and high-performance computing. Seeking a 6-month M2 internship in quantum information and quantum computing.

EXPERIENCE

THE STATE UNIVERSITY OF CAMPINAS (UNICAMP) *Campinas, Brazil*
Quantum Information Research Intern *05/2026 – Present*

- **Quantum Nonlocality & SVMs:** Engineered a novel SVM pipeline to classify Local Hidden States (LHS) and quantum steerability by compressing 2-qubit density matrices into a 9D Fano representation. Extracted a novel algebraic formula to capture nonlocality, overcame LP bottlenecks using MOSEK, and presented findings to UNICAMP physics department. **[Code]**
- **Multipartite Bell Scenarios** *(Paper in preparation)*: Defining high-dimensional polytopes and applying vertex annotation (PORTA, PANDA) to discover novel Bell inequalities. Modeling quantum correlations via convex optimisation (CVXPY), linear programming, and see-saw techniques.
- **Quantum Catalysis** *(Paper in preparation)*: Investigating catalytic power of quantum states across dimensions, analysing entanglement as a novel thermodynamic and informational resource.

CHARLES UNIVERSITY *Prague, Czech Republic*
AI Researcher – Bachelor's Research *2024 – 2025*

- **Adversarial AI:** Achieved $>$75% attack success rate on Vision Transformer and CNN architectures by engineering adversarial patch attacks using PyTorch. **[Code]** **[Thesis]**
- **HPC Simulation:** Designed and deployed reproducible Python-based machine learning pipelines, consuming over 500 GPU days on the MetaCentrum supercomputing cluster.
- **Model Optimisation:** Integrated pre-trained models and applied optimisation techniques to improve reliability and performance across large-scale image datasets.
- **Scientific Workflow:** Collaborated in an agile research team, leveraging Git for version control and systematically documenting experimental methodologies for academic publication.

EDUCATION

SORBONNE UNIVERSITY *Paris, France*
Master of Science in Quantum Information (QI) *2025 – Present*
Relevant Coursework: Quantum Algorithms, Quantum Dynamics, Quantum Information Theory, Mathematical Algorithms & Complexity, Data Science & Statistical Learning, Photonic Quantum Computing, Quantum Cryptography.

CHARLES UNIVERSITY *Prague, Czech Republic*
Bachelor of Science in Computer Science – Artificial Intelligence *2021 – 2025*
Thesis: *Adversarial Examples Against Vision Transformers* (Supervisor: Doc. Mgr. Martin Pilát, Ph.D.)
Relevant Coursework: Linear Algebra, Probability & Statistics, Graph Theory, Mathematical Analysis, Machine Learning, Computer Vision, Natural Language Processing.

INITIATIVES

LONG SWIM TO FREEDOM *France & South Africa*
Co-Founder & Web Developer *2026 – Present*

- Co-founded non-profit initiative dedicated to rural child drowning prevention and rhino anti-poaching around the iSimangaliso Wetland Park.
- Designed and developed the campaign website to mobilise international awareness, secure partnerships, and facilitate donations for community swimming programmes and ranger units.

SKILLS

Quantum Computing: Quantum Algorithms, Quantum Information Theory, Hybrid Quantum-Classical Algorithms, Quantum Cryptography
Mathematical Optimisation: Convex Optimisation, Linear Programming, Semidefinite Programming (SDP), MOSEK, CVXPY, PORTA, PANDA, NumPy, Pandas, SciPy
Machine Learning: PyTorch, Hugging Face Transformers, Scikit-learn, Vision Transformers (ViTs), Adversarial Robustness, Computer Vision
Programming Languages: Python (Advanced), Julia, C#, Java, Haskell, MATLAB, Bash
Software & Infrastructure: Git/GitHub, Docker, Singularity (Containerisation), Linux, High-Performance Computing (HPC Workflows)
Spoken Languages: English (Fluent), Georgian (Native), Russian (Intermediate), French (Beginner)